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Simple reciprocal electric circuit exhibiting exceptional point of degeneracy

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Abstract

An exceptional point of degeneracy (EPD) occurs when both the eigenvalues and the corresponding eigenvectors of a square matrix coincide and the matrix has a nontrivial Jordan block structure. It is not easy to achieve an EPD exactly. In our prior studies, we synthesized simple conservative (lossless) circuits with evolution matrices featuring EPDs by using two LC loops coupled by a gyrator. In this paper, we advance even a simpler circuit with an EPD consisting of only two LC loops with one capacitor shared. Consequently, this circuit involves only four elements and it is perfectly reciprocal. The shared capacitance and parallel inductance are negative with values determined by explicit formulas which lead to EPD. This circuit can have the same Jordan canonical form as the nonreciprocal circuit we introduced before. This implies that the Jordan canonical form does not necessarily manifest systems' nonreciprocity. It is natural to ask how nonreciprocity is manifested in the system's spectral data. Our analysis of this issue shows that nonreciprocity is manifested explicitly in: (i) the circuit Lagrangian and (ii) the breakdown of certain symmetries in the set of eigenmodes. All our significant theoretical findings were thoroughly tested and confirmed by numerical simulations using commercial circuit simulator software.

Keywords: electric circuit, exceptional point of degeneracy (EPD), Hamiltonian, Jordan canonical form, Lagrangian, nonreciprocity, sensor.

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